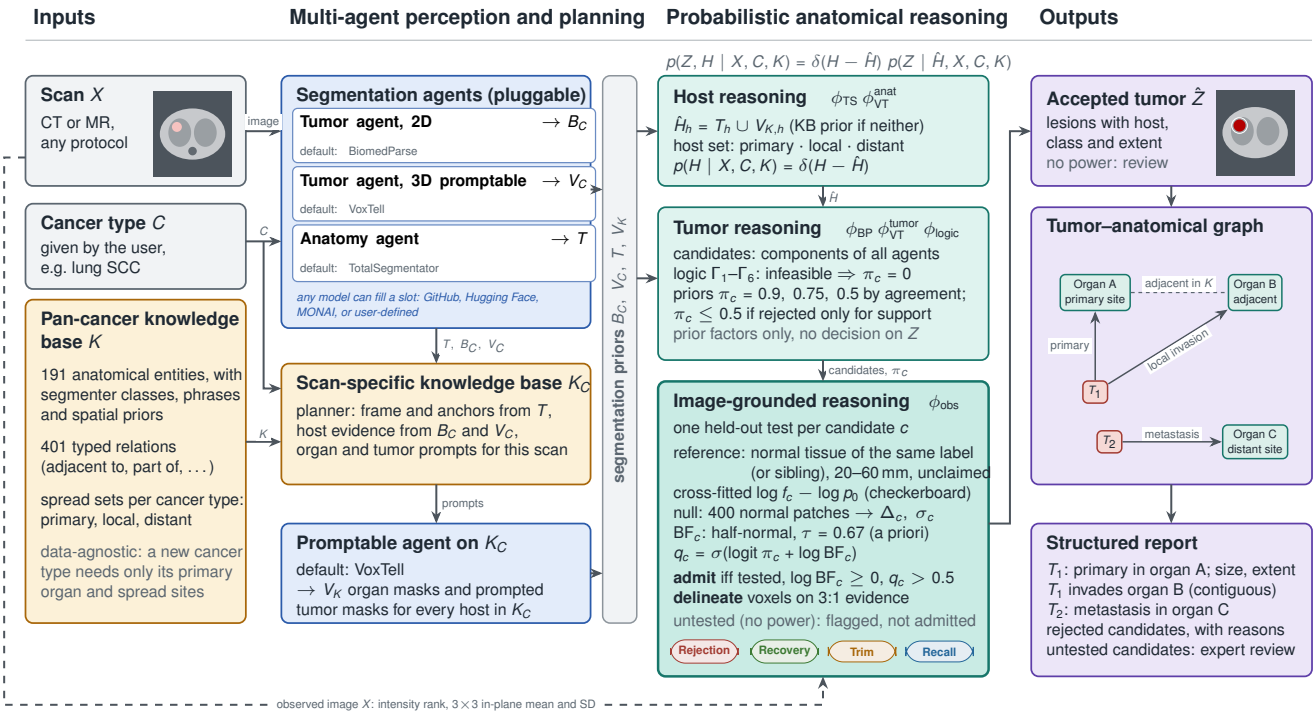


Figure 1: Pan-cancer multi-agent tumor segmentation and reasoning. Given a scan X and a user-given cancer type C, pluggable segmentation agents (defaults: BiomedParse for 2D tumor, VoxTell for 3D promptable tumor and anatomy, TotalSegmentator for anatomy; any model can fill a slot) produce tumor and anatomy masks. A planner turns the pan-cancer knowledge base K into a scan-specific prompt plan K_C. Set-based reasoning turns the agents' masks into claims, anchors, outlines and lesions, with each anchor's host candidates, support level and candidate extents. The host (normal model) is inferred probabilistically from anatomy and reported as a credible set of organs; given the host, image grounding tests whether the anchor is a tumor against likelihoods learned only from expert annotations, at a stated error level α, and the extent is chosen among the agents' outlines by a prior learned on expert data. [figure to be updated for v7/G1]

1 Introduction

- Build a pan-cancer multi-agent tumor segmentation system
- 2D, 3D tumor segmentation models, anatomical segmentation models could be any, even allow users to define or use their private models
- Beyond segmentation, provide anatomical graph and explanation

2 Methodology

Figure 1 summarizes the pipeline; the components are detailed below.

2.1 Conditional Probabilistic Anatomical Reasoning

We formulate tumor segmentation as conditional probabilistic anatomical reasoning that jointly infers each tumor, its host organ and its extent, by combining the outputs of several segmentation agents with anatomical knowledge, set-based logic and image evidence. Let X be the image, C the cancer type and K the data-agnostic anatomical knowledge base. The reasoning is organised so that early steps *generate* candidates and graded evidence and decide only what is impossible; competing explanations stay open until the image has been consulted, and decisions are taken last, by an explicit policy.

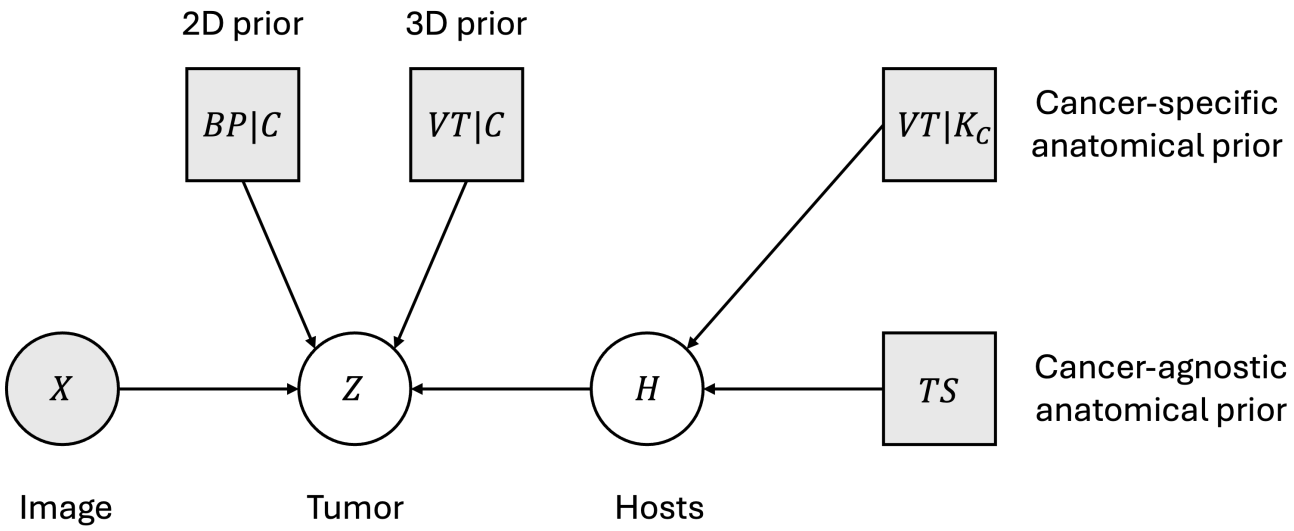


Figure 2: Conditional probabilistic anatomical reasoning. Each tumor candidate needs two kinds of reasoning, both grounded in the image: *host* (which organ it belongs to, and that organ's normal tissue) and *tumor* (whether it is a tumor and what its extent is). Host and tumor reasoning turn the agents' masks and the knowledge base into host candidates, support levels and candidate extents; image grounding supplies likelihoods learned from expert annotations; inference is conditional: the host and its normal tissue are inferred probabilistically first, and the tumor is tested against the expert likelihoods given them.

Segmentation agents. Agents are pluggable and treated equally. A *tumor agent* $k \in \mathcal{K}$ returns tumor masks for text prompts: one cancer-specific prompt (for example *kidney tumor* for KIRC) and, if the agent is promptable, one prompt per host of the cancer type in the field of view. An *organ agent* returns organ masks. With the default agents,

$$\begin{aligned} B_C &= f_{\mathrm{BP}}(X, C), \qquad V_C = f_{\mathrm{VT}}(X, C, K_C), \\ V_K &= f_{\mathrm{VT}}(X, K_C), \qquad T = f_{\mathrm{TS}}(X), \end{aligned} \tag{1}$$

where B_C and V_C are tumor masks (VT's cancer prompt and its host prompts from the plan K_C, Section 2.2), and V_K and T are organ masks. The agents currently available are listed in Table 1; the framework only assumes that a tumor agent returns a mask for a text prompt (or for the cancer type) and an anatomy agent returns labelled organ masks, so further models can be added without changing the reasoning.

We write $A_{k,o}$ for organ agent k's mask of organ o ($A_{\mathrm{TS},o} = T_o$, $A_{\mathrm{VT},o} = V_{K,o}$). Agreement counts agents, not prompts: different prompts of one agent are one rater. Every piece of support is stored with its provenance (which agent, which prompt), so that an agent confirming its own proposal, for example a VT tumor mask supported only by VT's organ mask, is not counted as independent confirmation.

Anatomy. Organ masks are not taken at face value. Each mask receives a confidence $\mu(A_{k,o}) \in \{0, \frac{1}{2}, 1\}$ from the organ–organ relations of the knowledge base (*host reasoning*, Section 2.3): 0 only for a corrupt or geometrically impossible mask, $\frac{1}{2}$ for a mask that violates a normal-anatomy expectation (which disease can cause), 1 otherwise. Disagreement between agents is kept rather than hidden: each organ is represented by three zones,

$$\hat{H}_o = \bigcap\nolimits_{k:\,\mu(A_{k,o})=\mu_o^{\max}} A_{k,o} \setminus \bigcup\nolimits_{o'} C_{o,o'}, \qquad B_o = \Big(\bigcup\nolimits_{k:\,\mu(A_{k,o})>0} A_{k,o}\Big) \setminus \big(\hat{H}_o \cup \bigcup\nolimits_{o'} C_{o,o'}\big), \tag{2}$$

the *core* $\hat{H}_o$, where the most reliable masks agree; the *uncertain boundary* B_o, claimed by some masks only; and the *contested* regions $C_{o,o'}$, where masks of two organs that the knowledge base does not relate by containment overlap. The territory of o is $\Theta_o = \hat{H}_o \cup B_o \cup \bigcup_{o'} C_{o,o'}$. Each organ the planner expects is in one of three states: *present* (some mask with $\mu > 0$), *unsegmented* (inside the field of view but without such a mask, because of agent failure, distortion or replacement by tumor) or *out of view*. An unsegmented organ is represented by a knowledge-base region (its own vertebral span, on its own side) with confidence $\frac{1}{2}$; an organ with less than half of its span in the field of view is out of view and is not a host. The anatomy $A = \{\hat{H}_o, B_o, \sigma_o\} \cup \{C_{o,o'}\}$, with σ_o the state, is a deterministic

Table 1: Pluggable segmentation agents. Any model that fills a role through the same interface can be added; the reasoning does not depend on which models are used.

Agent	Role	Dim.	Modalities	Input and output
BiomedParse [10]	tumor	2D	CT, MR (and 7 others)	text prompt (cancer-specific phrase); tumor mask per slice, stacked to 3D
VoxTell [8]	tumor and anatomy	3D	CT, MR, PET	free-text prompt (cancer phrase, host-specific tumor phrases, organ names and aliases from K_C); one mask per prompt
Other tumor models	tumor	2D/3D	–	any model returning a tumor mask for a text prompt or the cancer type
TotalSegmentator [9, 1]	anatomy	3D	CT, MR	automatic; labelled organ masks (CT and MR task models)
VoxTell [8]	anatomy	3D	CT, MR, PET	free-text organ prompts from the plan K_C (main phrase and aliases)
NV-Segment-CTMR [6, 3]	anatomy	3D	CT, MR	automatic; 345+ anatomical classes; VISTA3D architecture
Other anatomy models	anatomy	3D	–	any model returning labelled organ masks

function of the organ agents' outputs and K, and is *cut* from the tumor model [5, 7, 4]: tumor inference never reshapes an organ mask, and the normal tissue of each host is computed once from the anatomy and the agents' claims and is not updated. The knowledge base gives the cancer type's *expected host list* $\mathcal{H}_C = \{(h, \kappa_h)\}$ with the role κ_h of each organ: the primary organ, organs at risk of local invasion and distant sites (Section 2.2). The list decides which organs are expected hosts; it does not weight them, and other organs can join a lesion's candidates when the anatomy demands it (Section 2.3). Roles enter inference only through one warning signal, the missing site agreement of a distant host, and through invasion explanations between the primary organ and its local-invasion organs (Section 2.3); otherwise they are attached to a host after inference.

Set-based units. Reasoning acts on sets produced by the agents, never on redrawn voxels.

- A *claim* is a connected component of one agent's tumor mask from one prompt. It is the smallest unit and is never cut.
- An *anchor* a is a located core: a connected component of the voxels claimed by at least two different agents (*two-agent anchor*), or a component of one agent's claims that overlaps no two-agent anchor (*single-agent anchor*).
- The *outlines* Ω_a of an anchor are the claims that overlap it. Relative to a, an outline consists of the shared core, *extensions* (its remainder connected to a) and *lobes* (parts of its remainder joined to a only through a neck, Section 2.3). A lobe is an anchor of its own, so the support of a core is not transferred to everything connected to it.
- The *host candidates* $\mathcal{H}_L$ of a lesion are the expected hosts in view (present, or unsegmented and represented by a knowledge-base region), at any distance, together with the segmented organs that *join* the lesion's list because the anatomy points to them: an organ joins if it is closer to the lesion than every expected host, or holds the largest share of it (Section 2.3). Distance and overlap rank candidates; they never remove an expected host.
- A *lesion* L is a group of anchors within $\tau = 10\,\text{mm}$ of each other. Grouping is geometric and independent of the host; anchors of a lesion share one host.

Each anchor defines a *grounding case* $c_a = (a, \Omega_a, \mathcal{H}_{L_a})$: the anchor, its tumor candidates and the host candidates of its lesion.

Latent variables. For each anchor a we infer its host $h_a \in \mathcal{H}_{L_a}$, its existence $z_a \in \{0, 1\}$ (tumor or normal structure) and, if it is a tumor, its extent $\xi_a \in \mathcal{X}_a$, chosen from candidate extents built from its outlines by set operations,

$$\mathcal{X}_a = \Big\{ a,\ \ \omega\ (\omega \in \Omega_a),\ \ U_a = \textstyle\bigcup_{\omega \in \Omega_a} \omega,\ \ U_a \setminus \bigcup_{o \notin \mathcal{H}_{L_a}} \hat{H}_o \Big\}. \tag{3}$$

Outlines flagged by the anatomical tests (a leak, an organ overlap) remain candidates with a lower prior, because an apparent leak may be invasion and an organ overlap may be replacement by tumor; only corrupt outputs are removed. The extents run from the agreed core a (maximal trimming), through each agent's reading, to the union of the readings and the union without its leaks into organs that are not hosts of a. In stage 1 the delineation is a choice among the agents' proposals, which locates each tumor robustly; it is refined voxel by voxel in stage 2 (Section 2.4). Existence and attribution are kept apart: the role of the host (primary, local invasion, distant, or an organ that joined the list) is reported with the tumor and does not change its decision.

Conditional probabilistic formulation. Given the anatomy, the anchors are the units of inference. Writing N for the normal model (the host and its normal tissue) and T for the tumor model (existence and extent), the target for each grounding case is

$$\boxed{p(T, N \mid c_a, X) = p(T \mid N, c_a, X)\; p(N \mid c_a, X)} \tag{4}$$

that is, the normal model is inferred first and the tumor model given it. The two factors are treated according to what expert annotations can estimate. The normal model of host h is the pair $(h, \mathcal{N}_h)$: the host and its normal tissue, the core $\hat{H}_h$ without any agent's claims (Section 2.3). It is inferred probabilistically, from anatomy alone,

$$p(N = h \mid c_a, X) = P(h \mid L_a) = \frac{\exp\big(\beta^\top \phi_h(L_a)\big)}{\sum_{h' \in \mathcal{H}_{L_a}} \exp\big(\beta^\top \phi_{h'}(L_a)\big)}, \tag{5}$$

a choice model over the host candidates of the anchor's lesion L_a, with anatomical compatibility features ϕ_h (overlap with the territory, boundary contact, relative position and mask confidence; Eq. (8)) and weights β fitted on expert tumors whose organ of origin is known. Every candidate enters with the same weight: the cancer type decides which organs are candidates, not which one is preferred. The site of a tumor often cannot be settled from the image alone, and difficult cases are resolved by pathology, so the host is reported as a *host credible set* $\mathcal{S}_a$, the smallest set of candidates whose probability is at least $1 - \alpha$, rather than as a single organ.

The tumor model given the normal model is not given a posterior probability. Its prior, how often an agent's candidate is a tumor, depends on the agents, and the agents are not run on expert data, so it cannot be estimated. Existence is therefore *tested* against expert likelihoods given the host, at an error level α that plays the role of the significance level of a statistical test: the error rate we accept. The observation is the existence evidence $E_a^{\mathrm{ex}}(h)$ of the anchor under host h, with likelihood ratio

$$\Lambda_a^{\mathrm{ex}}(h) = \frac{p\big(E_a^{\mathrm{ex}}(h) \mid T, h\big)}{p\big(E_a^{\mathrm{ex}}(h) \mid N, h\big)}. \tag{6}$$

An outcome *rules out* a tumor if at most a fraction α of expert tumors but more than a fraction α of expert normal structures show it, and *confirms* a tumor in the opposite case; otherwise it is neutral (Section 2.3). The agents' support enters as an ordinal *support level*, not as a probability (Section 2.3). The extent of a tumor is chosen among its candidate extents by an extent prior $w_a^0(\xi)$ learned entirely on expert tumors and their simulated agent outlines (Eq. (9)). The image does not re-rank the hosts: host-frame evidence is computed against each host's own normal tissue, so evidence under different hosts is evidence about different data.

Host and tumor reasoning. Both are set-based relational checks against the knowledge base. K stores organ–organ relations $\mathcal{R}_{AA}$ (containment: *has part*, *lies in*, *invested by*; *adjacent to* with contact and direction; spatial priors with direction and distance; and, by default, disjointness of organs it does not relate by containment) with unary organ attributes (volume range, number of components, side, vertebral span), and tumor–organ relations $\mathcal{R}_{TA}$ (inside, straddling or touching an organ, extending into an adjacent organ, lying at a cancer-specific distant site, being or containing an organ, lying in an organ that is not a host of the cancer type). *Host reasoning* evaluates $R(A_{k,o}, A_{k',o'}) \in \mathcal{R}_{AA}$ between anatomy masks, together with the unary checks; it gives the anatomy: mask confidences, organ states, cores, uncertain boundaries and contested regions. *Tumor reasoning* evaluates $R(u, \Theta_o) \in \mathcal{R}_{TA}$ between tumor units u (claims, anchors, outlines) and the anatomy; it gives the host candidates, the compatibility features $\phi_h(a)$, the anatomical flags, the support level ℓ_a and the extent prior $w_a^0(\xi)$. Host reasoning reads anatomy masks only, so the anatomy stays cut.

Candidates, not decisions. Host and tumor reasoning use the agents' masks and the knowledge base, never image intensities, and they do not decide the tumor. Only what is impossible is decided early: a corrupt agent output is removed and an organ out of view is not a host. Everything else that is anatomically *unsupported* (a

single agent, a flat claim, a host-prompt claim outside its host, a leaking or organ-sized outline, a distant lesion without site agreement) lowers the support level or flags a unit, and the image can overturn it. Silence of another agent is not evidence against a candidate, and not evidence for it. The cancer is confirmed and the scans precede surgery, so a primary tumor exists in the scan; this is represented at scan level (Section 2.3) and does not make any particular candidate near the primary organ the primary tumor.

Image grounding: evidence the agents do not use. A segmentation agent outlines what looks locally abnormal, so local contrast is what selected a candidate; it cannot distinguish a tumor from the normal structures an agent mistakes for one. Existence evidence therefore comes only from properties the agents do not model and that were found to transfer from expert data to unseen agent candidates: *context* (whether the candidate's own deviation from its host stays closed or continues into the host), *internal structure* (spatially structured heterogeneity in the candidate's own frame) and *geometry* (elongation). Contrast only routes: it decides whether the context test applies. Extent evidence asks a different question, whether the tissue an outline adds to the core looks like the core or like the host's normal tissue. The likelihoods of existence evidence are frequencies on expert annotations only,

$$\begin{aligned} p(E^{\text{ex}} \mid T, h) &: \text{ frequency over expert tumor masks and their partial cores,} \\ p(E^{\text{ex}} \mid N, h) &: \text{ frequency over expert normal anatomy of host } h, \end{aligned} \tag{7}$$

and the extent prior (its size and flag terms) is learned from expert tumors and their simulated agent outlines. No likelihood or prior is fitted on agent outputs or on the evaluated cohort. Evidence that cannot be computed is marginalised (it is not evidence).

Joint inference. Inference follows Eq. (4) and is computed once; there is no iteration and no output is fed back.

1. *Normal model.* For every lesion L, $P(h \mid L)$ (Eq. (5)) is computed on the lesion as a whole and shared by its anchors, together with its host credible set $\mathcal{S}_L$; for every candidate host, the normal tissue $\mathcal{N}_h$ is fixed. A host must be near its tumor: if every host of $\mathcal{S}_L$ lies farther than $\delta = 10\,\text{mm}$ from the lesion, the host is *unknown* and no decision above uncertain is taken.

2. *Tumor model given the normal model.* For every anchor and every host $h \in \mathcal{S}_{L_a}$, the existence evidence is tested (rules out, neutral or confirms) and combined with the anchor's support level into a decision (Table 2). The decision must be the same under every host of the credible set; otherwise the anchor is uncertain.

3. *Extent.* For an accepted anchor, the extent is $\hat{\xi}_a = \arg\max_{\xi \in \mathcal{X}_a} w_a^0(\xi)$.

The outputs are, for every anchor, the host probabilities and credible set, the support level, the test outcome and Λ^{ex} under each host of the credible set, the extent prior and the chosen extent, and the decision. The stage-1 tumor segmentation is $\hat{Z} = \bigcup_a \hat{\xi}_a$ over the accepted anchors, each reported with the role of its host. It localizes the tumors; their voxel-wise delineation is refined in stage 2 (Section 2.4).

2.2 Pan-cancer knowledge base and prompt planning

This section describes how the anatomical knowledge base K is built once for all cancer types, and how, for each scan, it is turned into the final VT prompt plan K_C of Eq. (1). K is a curated, image-independent graph; K_C is a finite list of prompt jobs, and $V_K = f_{\text{VT}}(X, K_C)$ is the stack of masks VT returns for these jobs. No component of K or of the planner uses patient sex, and no rule or constant is specific to an organ, a cancer type or a dataset.

Anatomical entities. K contains 204 anatomical entities. Each entity has a category (solid organ, hollow organ, gland, bone, vessel, lymph-node station, muscle, fascia, fat, duct, space, nerve, landmark or region) and a laterality (left, right, midline or bilateral). It also records how it can be observed: the TotalSegmentator classes that represent it for CT and for MR, if any; a VT main phrase with up to two aliases, each graded as in-vocabulary, near-vocabulary or out-of-vocabulary against the published VT training vocabulary; its craniocaudal span between vertebral anchors; whether it is always present, variably present or absent after common surgery; a typical volume range; and, for CT, a typical Hounsfield-unit range. Finally, each entity that can host a tumor carries a tumor phrase (for example *liver tumor*). Thirty-nine entities are *anchors*: structures that TotalSegmentator segments reliably and that therefore fix the coordinate frame of a scan (vertebrae, major vessels and large organs). Landmarks are tied to vertebral levels, so that every spatial prior in K is expressed relative to the patient's own skeleton rather than in millimetres.

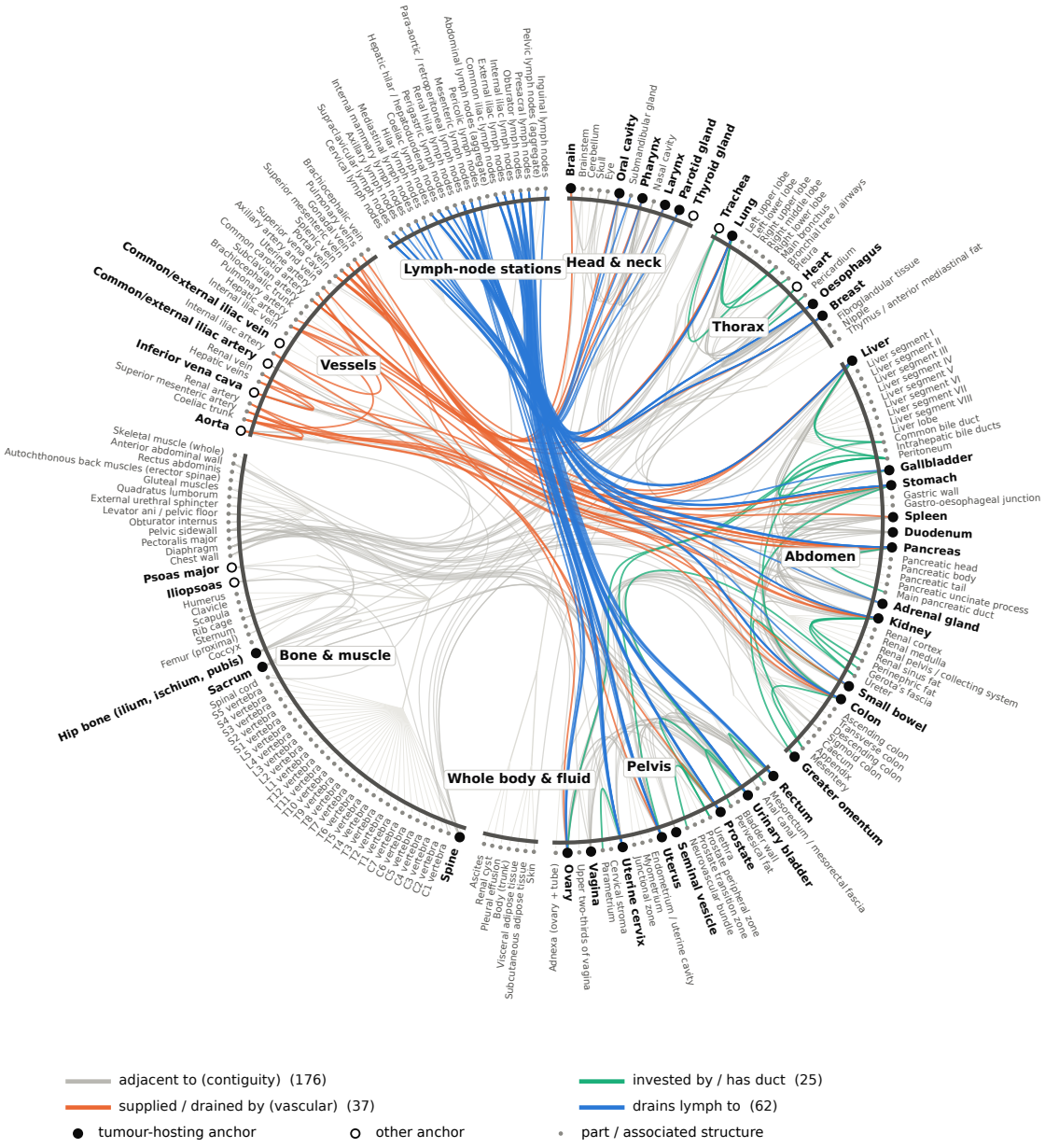


Figure 3: Pan-cancer anatomical knowledge graph

Anatomical relations. Entities are connected by 433 typed, directed relations. Ten relation kinds are defined, of which nine are used: *adjacent_to* (176), *has_part* (97), *drains_lymph_to* (62), *lies_in* (23), *supplied_by* (21), *invested_by* (16), *drained_by* (16), *landmark_for* (13) and *has_duct* (9); the tenth, *wall_of*, is reserved. Each relation kind has a fixed spatial gate and a default priority. The gate states where the target may lie relative to the source: inside it (*has_part*), in a shell around it (*invested_by*), in a neighbouring box or directional shell (*adjacent_to*), in a corridor along a vessel or duct (*supplied_by*, *drained_by*, *has_duct*) or in a body region (*drains_lymph_to*); *lies_in* and *landmark_for* carry no gate. Gates are later used to check VT outputs against structures that are already segmented, and priorities order prompts when the budget is exceeded. The relations encode the anatomy that staging systems refer to: direct extension into a neighbouring organ, invasion of an investing fascia or fat, and lymphatic drainage to a regional station.

Cancer-type spread sets. For each cancer type C, K stores a weighted host set $\mathcal{H}_C$, which is the host set used in Section 2.3. Only two items are specified per cancer type: the primary organ and a list of distant metastatic sites. Organs at risk of local invasion are not curated; they are derived automatically as the entities reached from the primary organ by an *adjacent_to* or *invested_by* relation that carries an invasion staging tag, with the relation's side rule for paired organs. A route that ends at a part no agent segments (for example the sigmoid colon) is mapped to its nearest segmented whole along *has_part* (the colon), with the same role; distant sites are mapped in the same way. The reasoning takes this list from K; the planner's use of the initial tumor priors (step iii below) orders prompts and never removes a host from it. Organs present in one sex only (uterus, cervix, vagina, ovary, prostate, seminal vesicle) are hosts only where K lists them for the cancer type: patient sex is not used, so a mask of such an organ cannot be verified and is never added by proximity. Each cancer type also has a short ensemble of tumor phrases (for example *lung cancer*, *lung tumor*, *lung mass*), and generic templates produce phrases for any host without a curated one. The distant-site lists are user-defined inputs rather than an exhaustive account of metastatic spread. Here we restrict each list to the most common metastatic sites reported for that cancer type that are also anchors of K, and cancer types without a curated list default to liver, lung, adrenal gland and spine. Head-and-neck anchors (brain, oral cavity, pharynx, larynx and parotid gland) are part of K but are in no default list, because the body scans considered here rarely include the head and neck. Sites that are not anchors, such as the pleura and distant lymph nodes, are not prompted. A user can extend a list, or add an anchor, without changing any other part of the framework, and only the affected cancer types need to be re-planned. The current base covers 17 TCGA cancer types, plus a CPTAC entry whose cancer type is resolved per case. Adding a new cancer type therefore requires only its primary organ, its metastatic sites and, optionally, its phrase ensemble; all other knowledge is shared.

Planning K_C for a scan. For a given scan, the planner builds K_C deterministically in a single pass, without calling any segmentation model. Its inputs are the modality, the cancer type C, a summary of the TotalSegmentator output T (which classes are present, their volumes and their extents), and the initial tumor priors B_C and V_C of Eq. (1). It proceeds in five steps.

(i) Frame. The vertebrae found in T define a craniocaudal ruler. The landmarks of K are placed on this ruler, and every entity span is converted to a slice range of the scan. This step also determines which entities lie inside the field of view.

(ii) Anchors. Each anchor is labelled *found* (present in T with a plausible volume and position), *expected* (inside the field of view but missing from T) or *missing* (outside the field of view). Found anchors give the reference structures against which the gates of later prompts are evaluated.

(iii) Tumor evidence. A component of B_C or V_C is *validated* when the two priors agree on it. For each host $h \in \mathcal{H}_C$ the planner records an evidence level $e_h \in \{0, 1, 2\}$: 2 if a validated component lies in or within a reach of 10 mm of h, 1 if a component from one prior does, and 0 otherwise. It also records the side of the primary tumor for paired organs. The planning priority of the host is $r_h(1 + e_h)$, where r_h ranks its role (primary 1, local invasion 0.3, distant 0.15); it only orders prompts under the budget and plays no part in inference. The initial tumor priors therefore influence only the order and the extent of the anatomical prompts; they never remove a host from $\mathcal{H}_C$.

(iv) Prompt tiers. Prompts are generated in four tiers. Tier 0 adds ruler prompts (spine, sacrum, hip bone) only when T provides neither vertebrae nor fallback landmarks, so that a frame can always be placed. Tier 1 completes the anchors that are expected but missing from T, each gated by its spatial prior. Tier 2 adds *anatomical profiles* by walking the relation graph: two hops from the primary organ, one hop from each secondary host, and the structures that can be reached from any host with tumor evidence $e_h > 0$. Tier T contains the tumor prompts: one prompt with the tumor phrase of each host of $\mathcal{H}_C$ inside the field of view. Tier T is issued for every such host regardless of the evidence e_h, so that a lesion missed by both initial priors can still be proposed. For the same reason, when the

initial VT tumor mask is empty or smaller than 0.5 ml, the cancer-type phrase ensemble is added for the primary organ. This is a wording ensemble for the same model, not a second rater: the initial and prompted VT outputs are counted as one model in tumor reasoning.

(v) Budget. The plan is limited to 80 prompts. Tiers 0, 1 and T are protected; only tier 2 is trimmed, lowest relation priority first, with items derived from a host given a one-step priority boost.

The output K_C is the list of prompt jobs that survive the budget. Each job records the entity, its side, the phrases to issue (the main phrase and its first two aliases), its tier, its gate, and, for tumor prompts, the host and its role. VT is run once per scan on the full volume with all jobs of K_C; there is no re-planning after inference. For each job, the masks of its three phrases are fused by voxel-wise majority (at least two of three). The gate is then applied as an output filter: each connected component of the fused mask is kept only when at least 80% of its voxels lie inside the region allowed by the gate. A completion structure that is empty after filtering is recorded as absent when K marks it as variable or removable by surgery, and as not found otherwise. The kept organ masks form V_K, which enters the organ references $\hat{H}_o$. The masks of the tier-T jobs are tumor proposals; they are added to VT's tumor masks V_C as host-prompt claims, each tagged with its prompted host, and are subject to the same host, tumor and image reasoning as every other claim.

Two properties of this construction matter for the reasoning that follows. First, although K is independent of the image, K_C is not: it depends on T through the frame and the anchors, and on the initial tumor priors through the evidence levels. This dependence is restricted to which structures are prompted and in what order, so it changes the coverage of V_K but not the meaning of any mask. Second, because tier T prompts every plausible host whether or not it contains a tumor, VT is regularly asked to segment a tumor that is absent. Its output for such a prompt can be a false positive inside the prompted host or elsewhere in the scan. These proposals are therefore treated as priors only, and whether they are accepted is decided by image grounding (Section 2.3).

2.3 Reasoning details

All operations are per scan on the native voxel grid. Distances are Euclidean in millimetres. The *acquisition axis* is the axis with the largest voxel spacing when the ratio of largest to smallest spacing is at least 1.5, and the superior–inferior axis otherwise; slice-wise operations use the plane orthogonal to it. No rule names a cancer type or an agent, no constant is tuned on clinical endpoints, and patient sex is never used. The source of every constant (expert data, knowledge base, or design choice) is listed in Supplementary Table S1; design constants are few and their sensitivity is reported.

Order of computation. (0) image quality control; (1) host reasoning: mask confidence, organ states, cores, boundaries and contested regions; (2) claims, anchors, lobes and outlines; (3) host candidates, anatomical compatibility and lesions; (4) anatomical flags, support levels and extent prior; (5) normal tissue and image evidence; (6) one-pass conditional inference; (7) decision policy.

Image quality control. Image grounding needs an image in which the anatomy agents can delineate normal organs. Each scan receives a state $Q \in \{\text{good}, \text{degraded}, \text{exclude}\}$ before reasoning. Let $\mathcal{E}$ be the organs the planner expects in view (obligatory presence, at least half of its vertebral span in the frame). An expected organ *fails* if an agent segmented it but every mask of it is infeasible ($\mu = 0$). An organ that no agent segmented does not count and leaves the denominator, because absence usually reflects the field of view or the protocol (for example breast MR) rather than image quality. Conflicts between organ agents shape the anatomy but do not grade the scan, because they mostly reflect one agent's error rather than the image. Failures are split by the tumor agents' agreement (voxels claimed by at least two agents' cancer prompts): an organ whose envelope it touches may have been replaced or swallowed by the tumor, which is a tumor sign, not a quality sign. With f the share of failed organs among the segmented expected organs untouched by tumor claims (computed when there are at least 4), $f \geq 0.25$ gives *degraded*. Missing anatomy alone never excludes a scan, because dedicated protocols (for example breast MR) show little anatomy besides the host; a scan is *excluded* only when $f \geq 0.5$ and an image measure fails as well. A scan is also *degraded* if its slice spacing exceeds 8 mm or the noise in a homogeneous reference tissue (paraspinal muscle on the common 5 mm grid, in-plane 1.5 mm; standard deviation in HU on CT, signal-to-noise ratio on MR) is beyond the 99th (CT) or 1st (MR) percentile of the expert datasets (76.6 HU and 1.64). Excluded scans leave the cohort and are reported with their reason; degraded scans are processed by host and tumor reasoning with image grounding switched off ($\Lambda = 1$).

Host reasoning. *Organ masks.* $A_{\mathrm{TS},o} = T_o$; $A_{\mathrm{VT},o} = V_{K,o}$ is VT's organ mask (main phrase, or the voxel majority of the main phrase and its aliases). Each mask is checked separately, so one agent's error does not condemn the organ, and no check removes an organ: checks set the confidence μ of a mask.

Unary relations. A mask has $\mu = 0$ only if it is corrupt (empty, or more than half of the field of view) or geometrically impossible: a paired organ more than 10 mm on the wrong side of the mid-sagittal plane, or a centroid outside the organ's vertebral span by more than the span's variability plus one level, both checked only when the midline, the header handedness and the vertebral levels are reliable. A mask has $\mu = \frac{1}{2}$ if it violates a normal-anatomy expectation that disease can cause, for every organ alike: volume above $2\,v_{\max}$ or below $0.5\,v_{\min}$ of the knowledge-base range (when the organ is fully in the frame), more components than the organ has, or cut by the field of view.

Pairwise relations. For organs with a boundary of their own (organs, glands, bones, muscles, vessels; not lymph stations, fat, fascia or spaces) that K does not relate by containment, the overlap of their masks (any agents) is the contested region $C_{o,o'}$; above 5% of the smaller mask both masks are recorded as conflicting. A part that lies less than 90% inside its whole, organs that K says abut but lie more than 10 mm apart, and spatial priors violated by more than their stated distance plus 10 mm are recorded as flags and lower the confidence of the masks involved to $\frac{1}{2}$; no mask is clipped or removed, because tumors displace and deform organs and either mask may be the wrong one.

Organ zones and states. The core, boundary and contested regions follow Eq. (2), where $\mu_o^{\max}$ is the highest confidence among o's masks. An organ is *present* if some mask has $\mu > 0$, *unsegmented* if it is expected in view (planner state *expected*) without such a mask, and *out of view* otherwise. An unsegmented organ is represented by its knowledge-base region: the slab between the vertebral levels of the organ's own span in K, widened by the span's variability plus half a level, placed on the patient's segmented vertebrae (on the planner's frame when fewer than two are segmented), restricted to the body (the union of all segmented structures) and, for a lateral organ, to its side of the midline; within the slab, the region is narrowed by K's directional relations to segmented landmarks (each relation is a cone of half-angle 45° with a 10 mm margin and a reach of 50 mm, and a landmark shadows what lies behind it; these settings are fitted on TotalSegmentator anatomy). If less than half of that span lies within the scan, the organ is out of view. The region says only where the organ could be: its core is empty, its territory is the region, and its organ confidence is $\frac{1}{2}$. The confidence of an organ, μ_o, is $\mu_o^{\max}$, or $\frac{1}{2}$ when unsegmented.

Anatomy-insufficient scans. Host reasoning needs at least one organ that is segmented confidently. If no organ of the scan has $\mu_o = 1$ (dedicated small-field protocols such as breast or prostate MR), the scan is *anatomy insufficient*: the host is assumed to be the primary organ (side undetermined), no host inference is made, the anatomical warning signals that need a host mask are not evaluated, and existence is tested with the tables without context. The record is flagged so that the assumed host is never mistaken for an inferred one.

Tumor reasoning. *Claims.* A claim is a 26-connected component of one agent's tumor mask from one prompt, of at least 0.5 ml; smaller components touching the envelope of a host of $\mathcal{H}_C$ are grouped per agent and prompt (≤ 5 mm apart) and kept if the group holds at least 0.2 ml. Claims are never dropped by shape or prompt. Two warning signals are recorded instead: *prompt mismatch* (a host-prompt claim with less than 50% inside the envelope of its prompted host) and *flatness* (through-plane extent at most 0.25 of the in-plane diagonal, with a median per-slice coverage of the host organ it overlaps most of at least 40%). Both lower the prior (below), because thin, surface-spreading, partially covered, exophytic or invasive tumors can produce them.

Anchors, lobes and outlines. With M_k the union of agent k's claims, the agreement set is the voxels in at least two different M_k. Its components of at least 0.5 ml are two-agent anchors; components of M_k that overlap no two-agent anchor are single-agent anchors. Every claim is an outline of each anchor it overlaps. For an outline ω of a, its remainder $\omega \setminus a$ is opened with a 3 mm ball; a component of the opened remainder that is connected to a in the remainder only through voxels removed by the opening, and holds at least 0.5 ml, is a *lobe*: it becomes a single-agent anchor of its own and is removed from ω as an outline of a. The rest of the remainder is an *extension* of a. A bridge of erroneous tissue can therefore neither lend a core's support to a distant region nor hide a separate lesion.

Host candidates and compatibility. Distances are shortest surface-to-surface distances, in millimetres with the true voxel spacing, between the lesion (the union of its anchors) and an organ's territory Θ_o. The host candidates of a lesion are (i) every expected host of $\mathcal{H}_C$ in view, present or unsegmented, at any distance, and (ii) the organs that join it. An organ outside the list joins if it is a segmented anatomical anchor of K that can host a tumor, with $\mu_o \geq \frac{1}{2}$, and either it is strictly closer to the lesion than every expected host, or it holds the largest share of the lesion among all segmented organs, more than the share outside every organ (shares on slice-filled territories). Three rules keep joining organs plausible: an organ present in one sex only never joins (above); an organ that

anatomy QC found in the wrong place relative to its knowledge-base landmarks (not between them, or beyond a spatial prior) does not join; and a part never joins beside its whole. These rules use no fraction or distance chosen by hand. The envelope of an organ, $\mathcal{E}_o = \{v : d(v, \Theta_o) \le 10\,\text{mm}\}$, is used only by the warning signals below; it does not limit the host candidates. The compatibility of a with a candidate h is graded, not binary; it is described by the features

$$\phi_h(a) = \big(o_h^{\text{core}},\, o_h^{\Theta},\, s_h,\, \not\Vdash_{\text{inside}},\, \not\Vdash_{\text{straddle}},\, \not\Vdash_{\text{exophytic}},\, \mu_h,\, \not\Vdash_{\text{KB}},\, \log(1 + d_h)\big), \tag{8}$$

where o_h^{core} and o_h^{Θ} are the fractions of a inside $\hat{H}_h$ and Θ_h; s_h is the share of a's two-sided 3 mm boundary band (the 3 mm outside a, other anchors excluded, and its 3 mm inner rim) in Θ_h; the position is *inside* if $o_h^{\Theta} \ge 0.9$, *straddling* if $0.1 \le o_h^{\Theta} < 0.9$, and *exophytic* otherwise; μ_h is the organ confidence; $\not\Vdash_{\text{KB}}$ marks a host represented by its knowledge-base region; and d_h is the distance from the lesion to Θ_h, capped at 60 mm. The weights β of the choice model (Eq. (5)) are fitted on expert data, where the organ of origin of each tumor is known, by maximising the probability of the true organ among the candidates of each case: expert tumors with their true organ and its neighbours as candidates, perturbed organ masks (leaks, merges, missing parts, the tumor included in the organ, the true organ unsegmented). The host cases contain intraparenchymal tumors only; exophytic and straddling tumors and invasion are not represented, and synthetic invasion cases are reported separately (Section 3.4). The fit uses an L_2 penalty $\frac{1}{2}\|\beta\|^2$. For a lesion, $\phi_h(L)$ is computed on the union of its anchors. There is no separate no-host candidate: distance is a feature, and a host that is too far is handled by the tolerance below. An unsegmented organ is scored on its knowledge-base region with $\mu = \frac{1}{2}$ and $\not\Vdash_{\text{KB}} = 1$. The fit includes testbed cases in which the true host is unsegmented and represented by its knowledge-base region, with the regions of neighbouring organs as negatives, so that a region which only says where an organ could be is preferred only when the anchor's geometry favours it.

Local invasion. A tumor of the primary organ p that has invaded a neighbour n lies across both organs. When n is a local-invasion organ of p in K and the lesion lies within 3 mm of both, the pair is scored as one more explanation of origin p (p invading n), with features computed on the union of the two organs. The choice is over origins: each origin is scored by its best explanation, so the relation adds no probability mass, and the invaded organ is reported with the host.

Tolerance on host distance. After the credible set is final, its distance to the lesion is $d(\mathcal{S}_L) = \min_{h \in \mathcal{S}_L} d_h$. If $d(\mathcal{S}_L) > \delta = 10\,\text{mm}$ the host is unknown and the decision is at most uncertain. The tolerance allows one voxel of segmentation error for the organ and one for the tumor at the coarsest common slice thickness (5 mm); it is a design value. An unsegmented host lies at distance 0 when the lesion is inside its knowledge-base region, so the tolerance cannot exclude it.

Lesions. Two-agent anchors within $\tau = 10\,\text{mm}$ of each other are grouped into a lesion (with chaining); a single-agent or lobe anchor joins the nearest group within τ, if any, and never connects two groups, so a single-agent bridge cannot join two tumors. Grouping is fixed in millimetres and does not depend on slice spacing. Lesion roles are descriptive and assigned after inference from the role of the most probable host: a *primary-host tumor*, a tumor in an organ at risk of local invasion, or a *secondary-site candidate*. Contiguous extension of one tumor into a neighbouring organ is represented by the local-invasion explanations above. The index lesion is chosen by the decision policy, after every lesion has been scored.

Anatomical flags. An outline *is* an organ (volume ratio to the organ ≥ 0.6, median per-slice Dice ≥ 0.6, $\ge 80\%$ inside it) or *contains* it ($\ge 80\%$ of the organ inside, organ $\ge 50\%$ of its volume); an outline *leaks* if at least half of its volume beyond the anchor lies in organs that are not the anchor's host candidates; an anchor is *contested* if it touches a contested region $C_{h,o'}$ with $o' \in \mathcal{H}_C$ (both are then host candidates). Flags are soft: a flagged unit keeps its candidacy and its prior is lowered.

Support level. The agents' support for an anchor is an ordinal level, counted over agents with provenance: $\ell_a(h) = 2$ for a two-agent anchor and 1 for a single-agent or lobe anchor, lowered by one, to no less than 0, for each warning signal that applies under host h: prompt mismatch and flatness (when less than half of the anchor is covered by its agents' claims without that signal), no confirmation (a single-agent anchor with another agent's raw output in h's envelope but not within 3 mm of a), and, for a distant host, no site agreement (less than 0.5 ml of the lesion claimed both by a prompt for h and by a different agent). Silence (no output of another agent anywhere in h's envelope) is not a signal: it removes the penalty for missing confirmation and adds no support. The levels assume only that the agreement of two agents is stronger support than one agent, and that a warning signal weakens support by one level. No probability is attached to them, because the agents are not run on expert data and their reliability cannot be estimated there.

Extent prior. Over $\mathcal{X}_a$, after merging extents with Dice ≥ 0.95,

$$w_a^0(\xi) \;\propto\; \psi\big(\log \tfrac{|\xi|}{|a|}\big) \prod_{f \in F(\xi)} \eta_f, \tag{9}$$

where $F(\xi)$ is the set of flags of ξ (leak and organ overlap, tested on ξ itself, and the prompt mismatch and flatness of the outlines that form it); ψ is a parsimony term, a piecewise-constant density ratio of the log volume ratio, $p(\cdot \mid \xi \text{ is the extent closest to the expert mask})/p(\cdot \mid \xi \in \mathcal{X}_a)$; and η_f is the corresponding factor of flag f, constrained to $\eta_f \leq 1$ because a flag lowers the prior. Both are estimated jointly, as a conditional logit over each case's extents with a small L_2 penalty (0.01), on simulated agent cores and outlines of expert tumors, holding out one collection at a time; flags without testbed examples (prompt mismatch, flatness) take the smallest fitted factor (Section 3.4; Supplementary Table S1). A large or flagged extent is therefore chosen only as often as such extents are closest to the expert masks. In the current version the flag factors are set to $\eta_f = 1$ and the prior is ψ alone, because the fitted factors depended on how the simulated outlines were constructed.

Image grounding. *Why stage 1 chooses an extent.* On expert tumor masks, a voxel classifier on the host-normalised intensity alone, told the true tumor ±5 mm, agrees with the experts only at Dice 0.67–0.85: half of kidney and pancreas tumors are iso-intense to their organ, the intensity edge is a 6–7 mm ramp, and experts draw about 2 mm outside its midpoint. Host-normalised intensity is therefore not enough to redraw a boundary, and stage 1 chooses among the agents' outlines. That choice inherits the agents' under- and over-segmentation, which stage 2 addresses with the patient's own tumor tissue, image edges and the credible hosts (Section 2.4).

Normal tissue of a host. $\mathcal{N}_h = \hat{H}_h \setminus \big(\bigcup_k M_k \oplus 2\,\text{mm}\big)$: the core of h without any agent's tumor claim, with a 2 mm margin; claims flagged as *being* an organ are not excluded, because an organ-sized outline is the organ rather than a tumor proposal, and excluding it would delete the reference. It is computed once and not updated, so no candidate, true or false, is absorbed into the reference that judges it. Depth is the distance below the surface of $\hat{H}_h \cup \xi$. In depth bins $(0, 1.5, 3, 5, 8, 12, 18, 25, 35, 50\,\text{mm}$ and beyond), the median and robust standard deviation of the image smoothed by a 1 mm Gaussian (in-plane when slices are thick) over $\mathcal{N}_h$ define the depth-standardised deviation $Z(v)$ of every voxel from host h. The reference is valid if it holds at least 200 voxels within 50 mm of the anchor; otherwise the core without the anchors only is used (flagged), and if that also fails every host-frame sign is unknown. An unsegmented host has no normal tissue and uses the evidence without context.

Existence evidence. For anchor a under host h, each sign is yes, no or unknown:

- *Router* S_0: $\zeta = \text{median}_{v \in a}\, Z(v)$; the anchor deviates if $|\zeta| \geq 1.5$.
- *Context* S_1 (closedness): the ratio of the volume of a's same-sign deviation ($\text{sign}(\zeta)Z > 1.5$) that stays connected to a within 30 mm, inside Θ_h dilated by 2 mm, to the volume of a. Closed if ≤ 1.42, open if > 5.27. Applies when a deviates and at least half of a lies inside Θ_h dilated by 2 mm.
- *Internal structure* S_2: the median local standard deviation (2 mm) of the image standardised by a's own median and MAD, over a's core. Patchy if ≤ 1.047. Applies to anchors of at least 3 ml.
- *Elongation* S_3: $\sqrt{\lambda_{\min}/\lambda_{\max}}$ of the voxel coordinates. Elongated if < 0.363.

The thresholds are percentiles of expert data, fixed before any evaluation: closed and patchy at the 10th percentile of salient normal structures, open and elongated at the 90th and 10th percentiles of expert tumors. Local contrast is not evidence. Signs that did not transfer from expert data to agent candidates are not used (curvature-based vessel signs, rim–core profiles, appearance-density tests of existence and novelty scores). Organ overlap is not a sign: it enters once, as a flag of the extent prior. The signs are combined hierarchically into one outcome, contrast routing (S_0), then context (S_1), internal structure (S_2) and geometry (S_3):

$$\begin{aligned} \text{tumor-like} &\iff (\text{closed} \vee \text{patchy}) \wedge \neg\text{open} \wedge \neg\text{elongated}, \\ \text{organ-like} &\iff (\text{open} \vee \text{elongated}) \wedge \neg\text{closed} \wedge \neg\text{patchy}, \end{aligned} \tag{10}$$

and uncertain otherwise, including when the context test was skipped and elongation is the only organ-like sign (an iso-intense elongated tumor). The existence evidence $E_a^{\text{ex}}(h)$ is this outcome.

Existence likelihoods. $p(E^{\text{ex}} \mid T, h)$ is the frequency of the outcome over expert tumor masks of the public R2Seg collections and their partial cores (random sub-regions covering 30–70%, and three-slice cuts), because anchors are cores; dilated and leaking variants are not used here, as they concern the extent. $p(E^{\text{ex}} \mid N, h)$ is the frequency over expert normal structures of h's organ class (solid or hollow) in the public TotalSegmentator data that an agent can mistake for a tumor: salient intra-organ structures, vessel and bowel tube segments, organ edges, organ parts, adjacent organs, fluid and whole organs. A lesion whose host is unknown or assumed (anatomy-insufficient scans) uses the outcome without the context level. Signs are computed on a common through-plane grid of about 5 mm (slices along the superior–inferior axis averaged in blocks of $k = \lfloor 5/s + 0.5 \rfloor$, s the slice spacing; labels by majority vote), because thicker slices change closedness and internal structure of tumors and normal structures alike. Native

Table 2: Decision from the support level ℓ and the existence test at level α.

Support level	Confirms	Neutral	Rules out
$\ell = 2$ (two agents, no warning signal)	accept	accept	uncertain
$\ell = 1$	accept	uncertain	reject
$\ell = 0$	uncertain	reject	reject

resolution was considered for anchors smaller than 3 ml on thin-slice scans, where block averaging could blur small lesions, but on held-out expert data it separated small tumors from small normal structures no better than the common grid (Section 3.4), so the common grid is used throughout. Tables are built per organ class, size band (< 3, 3–20, > 20 ml) and modality (CT or MR). A stratum with fewer than 30 tumor or normal regions falls back, for both tables together, to the band within the modality, then to coarser strata, so the two tables always compare the same stratum. Normal scans of patients younger than 18 are excluded. Each class's frequencies are mixed with 10% of the pooled frequencies of the stratum, so a cell empty in both classes is neutral, and the log ratio of a cell supported by n regions is shrunk by $n/(n + 10)$. The smoothing constants are chosen by held-out likelihood on expert data. Unknown signs are marginalised over both tables.

Existence test. In each stratum, an outcome *o rules out* a tumor if $p(o \mid T, h) \leq \alpha < p(o \mid N, h)$, and *confirms* a tumor if $p(o \mid N, h) \leq \alpha < p(o \mid T, h)$: an outcome rare in both classes is not informative, and the condition implies the sign of Λ^{ex}. Every other outcome is neutral, and so is an anchor on a degraded scan. A confirmation counts only for an anchor that lies inside its host (at least half within Θ_h dilated by 2 mm), because internal structure alone lent support to normal structures outside the organ in held-out expert data. By construction, at most a fraction α of expert tumors are ruled out and at most a fraction α of expert normal structures are confirmed in each stratum; held-out collections show whether these rates transfer (Section 3.4). We use $\alpha = 0.05$ and report 0.01 and 0.10. On expert data, ruling out is the stronger test: few outcomes confirm a tumor of 3 ml or more, because normal structures that agents mistake for tumors often look tumor-like.

Extent evidence. An image term for the extent, which scores whether the tissue an outline adds to the anchor looks like the anchor or like the host's normal tissue on a region common to all extents, did not improve the choice of extent over the extent prior on held-out expert data; it is not used and is reported as an ablation (Section 3.8).

Decision policy. Decisions combine the support level with the existence test (Table 2). A conflict between them gives *uncertain*: the output never rejects an anchor that two agents agree on, nor accepts a contradicted one, without a person seeing it. An anchor is decided under every host of its credible set; if the decisions differ, it is uncertain. Uncertain anchors are reported and not segmented in stage 1; stage 2 refines both accepted and uncertain lesions. An anchor whose host is unknown (beyond the tolerance δ) is at most uncertain. If no anchor whose credible set contains the primary organ is accepted, the scan is flagged *primary not detected*, and the lesion whose credible set contains the primary organ and whose host is not unknown, with the highest support level (then the largest volume), is added to the output as a flagged override, with its anchors that are not rejected; the override is reported separately, with both its inferred and its imposed host. The index lesion is the largest accepted primary-host tumor, measured in the acquisition plane.

Output. Lesions keep their identity: a lesion's tumor is the union of the chosen extents of its accepted anchors, and a tumor is counted once. For every anchor the outputs are the host probabilities and credible set, the support level, the test outcome and Λ^{ex} under each host of the credible set, the extent prior and the chosen extent, the decision and any override, and the flags (missing normal reference, skipped context test, unsegmented host).

2.4 Stage 2: voxel-wise Bayesian tumor segmentation

From localization to segmentation. The two stages answer different questions. Stage 1 is *localization*. A tumor agent can tell that a region looks abnormal but not where it lies anatomically, nor whether that location makes sense for the cancer; this needs the anatomy agents and the knowledge base. Stage 1 therefore reasons globally, over relations between tumor units and organs, and outputs for every lesion L a coarse decision (accepted, uncertain or rejected), a set of anchors and candidate extents, and a host credible set $\mathcal{S}_L$ with probabilities $P(h \mid L)$. Stage 2 is *segmentation*. It works locally, voxel by voxel, on every accepted and uncertain lesion: where exactly the tumor is, which voxels the agents included wrongly or missed, and how the tumor is composed. The agents recognise tumor as a whole; they do not delineate its internal structure.

Between the stages the role of the host changes. In stage 1 the host is unknown and inferred (Eq. (5)); in stage 2 it is *known with uncertainty*: $P(h \mid L)$, $h \in \mathcal{S}_L$, renormalised over the credible set, is a fixed prior, and the anatomy stays cut. Stage 1 does not fix the tumor at voxel level. Its extents are proposals drawn by the agents, and its

decision concerns the lesion as a unit. The tumor mask is therefore a latent variable of stage 2, with a prior learned from expert data, and the stage-1 extent serves only as initialisation and as one input to that prior.

Region of interest. The model is defined on a region combining a local neighbourhood of the lesion with all of its credible hosts, because the host is uncertain and a tumor may extend into or arise from any of them:

$$R_L = \Big(\bigcup_{a\in L} a\Big) \oplus r \;\cup\; \bigcup_{h\in\mathcal{S}_L} \tilde{\Theta}_h, \tag{11}$$

restricted to the body. $\tilde{\Theta}_h$ is the territory of h, joined with its knowledge-base region when h is unsegmented or its mask is below the knowledge-base volume range (an organ largely replaced by tumor). The radius r is the 95th percentile, over expert tumors, of the distance from a simulated partial core to the farthest tumor voxel. If the posterior tumor reaches the boundary of R_L inside the body, R_L is enlarged by r around it and inference is repeated, so that a tumor much larger than its anchor is not cut by the region.

Generative model. Let y_v be the image features of voxel $v \in R_L$: the intensity, smoothed in the acquisition plane, and local means and standard deviations at two scales. Each voxel carries a latent label $Z_v \in \{0, 1, \dots, K\}$: 0 is background, $1, \dots, K$ are tumor components (for example enhancing, necrotic or cystic tissue). The tumor indicator is $t_v = \mathbf{1}[Z_v \geq 1]$, and the tumor mask is $\{v : t_v = 1\}$.

Spatial tumor prior. Tumors are not spatially homogeneous: they arise in particular organs, extend beyond the agents' outlines by characteristic distances, invade neighbours and respect some boundaries. A homogeneous label prior, as used for organ segmentation, does not express this. The prior probability that a voxel is tumor is

$$\pi_v = P(t_v = 1) = \sum_{h\in\mathcal{S}_L} P(h \mid L)\, \sigma\big(\eta^\top \psi_v(h)\big), \tag{12}$$

a mixture over the credible hosts. The features $\psi_v(h)$ are:

- the signed distance of v to the stage-1 extent $\hat{\xi}_L$;
- the stage-1 extent marginal $q_v = \sum_\xi w^0(\xi)\, \mathbf{1}[v \in \xi]$;
- whether v lies in the territory of h, in another credible host, or in a confidently segmented organ outside $\mathcal{S}_L$;
- the depth of v below the surface of that organ.

The weights η are fitted on expert data only: expert tumors with simulated agent outlines (partial cores, dilated and leaking outlines, outlines merged with the organ, lost hosts) and expert organ masks. Eq. (12) is therefore the frequency with which an expert tumor occupies a voxel in that position relative to an agent outline and to the anatomy. Unlike the extent prior of stage 1, it gives positive probability beyond the agents' outlines, which is what allows recall.

Edge-aware spatial coupling. Neighbouring labels are coupled by a Potts term whose weights are latent, as in BayeSeg [2]:

$$p(Z \mid \omega) \;\propto\; \prod_v \pi_v^{t_v}(1-\pi_v)^{1-t_v} \prod_{v\sim w} \exp\big(-\omega_{vw}\, s_{vw}\, \mathbf{1}[t_v \neq t_w]\big), \qquad \omega_{vw} \sim \mathrm{Gamma}(a_\omega, b_\omega), \tag{13}$$

where s_{vw} scales a neighbour pair by its physical distance, so thick slices couple less. The weight ω_{vw} is inferred: it becomes small where the image has an edge, so the tumor boundary can follow real edges rather than a fixed smoothness. The hyperparameters (a_ω, b_ω) are estimated on expert tumor masks, from the distribution of label changes along image gradients.

Clamp. The two-agent core of the lesion, eroded by one voxel, is fixed to tumor ($t_v = 1$). This conditions stage 2 on the lesion being a tumor. Whether it is one remains the stage-1 decision, and no prior on tumor existence is needed.

Appearance. Tumor appearance and background appearance are latent and specific to the scan. Tumors are heterogeneous and long-tailed (necrosis, calcification, haemorrhage), so each component has a Student-t likelihood:

$$p(y_v \mid Z_v = k) = \mathrm{St}\big(y_v \mid \mu_k, \Lambda_k, \nu\big),\; k = 1..K; \qquad p(y_v \mid Z_v = 0) = \sum_{c\in\mathcal{C}_L} \lambda_{v,c}\, \mathrm{St}\big(y_v \mid \mu_c, \Lambda_c, \nu\big). \tag{14}$$

The background is a mixture over the credible hosts' normal tissue and the surrounding anatomy, $c \in \mathcal{C}_L$. Its location weights $\lambda_{v,c}$ come from the organ territories, and the weight of host h is scaled by $P(h \mid L)$. The uncertainty about the host therefore carries into the voxel posterior instead of being resolved first. The means and precisions have conjugate Normal–Gamma priors. Their hyperparameters are estimated on expert data, per organ class and modality: the contrast of expert tumors relative to their organ's normal tissue and its spread, and the variability of normal tissue. The component weights have a Dirichlet prior. The number of tumor components $K \leq K_{\max}$ is chosen per lesion by the evidence lower bound, with $K_{\max}$ the largest number supported on expert tumors. On the scan, the agents' voxels only initialise the appearance; it is inferred jointly with the mask.

Inference. The posterior $p(Z, \theta, \omega \mid y)$ is approximated by mean-field variational Bayes, $q(Z)\, q(\theta)\, q(\omega)$, with closed-form coordinate updates of the conjugate families [2]. The label posterior is initialised from the stage-1 extent and the appearance from the voxels it implies. The outputs are:

- the *voxel tumor posterior* $p_v = q(t_v = 1)$, given that the lesion is a tumor;
- the *component posteriors* $q(Z_v = k)$, an intra-tumoral substructure map;
- the *refined tumor* $\hat{Z}_L$, the voxels with $p_v \geq \frac{1}{2}$ connected to the clamped core.

Comparing $\hat{Z}_L$ with the stage-1 extent gives the *trimmed* voxels $\hat{\xi}_L \setminus \hat{Z}_L$ (false positives of the agents' outline) and the *recalled* voxels $\hat{Z}_L \setminus \hat{\xi}_L$ (tumor the agents missed), each with the host territories it lies in. Lesions whose refined tumors meet are merged and the merge is reported. Recall comes from the image and the spatial prior only: the agents' discarded proposals (claims, crumbs and lobes that stage 1 did not keep) are never re-admitted.

Effect on the stage-1 evidence. Decisions are left unchanged. For every lesion, the stage-1 existence evidence (Section 2.3) is recomputed on $\hat{Z}_L$ under each host of $\mathcal{S}_L$, with the same expert tables and level α, and reported next to the stage-1 outcome, together with $\bar{p}$ over the stage-1 extent and over $\hat{Z}_L$. Whether the refined evidence should feed back into decisions is decided after this comparison.

Estimation and validation on expert data. Every prior and hyperparameter of stage 2 comes from expert annotations, never from agent outputs or the evaluated cohort:

- the spatial prior η;
- the edge hyperparameters (a_ω, b_ω);
- the appearance hyperparameters, per organ class and modality;
- the degrees of freedom ν;
- $K_{\max}$ and the radius r.

The data are R2Seg images with expert tumor masks, organ masks from TotalSegmentator and R2Seg, and simulated agent outlines. Validation holds out R2Seg collections in turn. It reports, per organ class and modality, voxel precision and recall, Dice, the over- and under-segmented fractions and the calibration of p_v, against the stage-1 extent, the anchor alone and the union of the agents' outlines. Stage 2 replaces the stage-1 extent only for the organ classes and modalities in which it improves the held-out Dice; elsewhere the stage-1 extent is kept, and this is reported.

3 Experiments and Results

3.1 Data

We use the public TCGA radiology collections of The Cancer Imaging Archive (TCIA). The cohort contains 3,291 baseline-study series (1,407 CT, 1,884 MR) from 15 cancer types: BLCA, BRCA, CESC, COAD, ESCA, KICH, KIRC, KIRP, LIHC, LUAD, LUSC, OV, PRAD, STAD and UCEC. TCIA states that only the baseline study precedes surgery, and post-surgical studies cannot be identified, so all evaluations use the baseline study only. The stage-evaluation scope is restricted to baseline series with the primary organ in the field of view: 3,116 series from 942 patients. Scans graded *exclude* by image quality control (Section 2.3) are removed and reported with their

reason ([TBD] series); *degraded* scans ([TBD]) are kept without image grounding. Clinical stage comes from the TCGA clinical records. No voxel-level tumour annotation exists for this cohort, so TCGA is used only for visual inspection, blind ratings of candidates and the stage-free yardsticks below; no likelihood, prior or threshold is estimated on it. Outputs of a pilot of 152 TCGA series were inspected visually for implementation errors only. The corrections that followed (the anatomy checks for small or multi-component organ masks, the laterality check, the grading of dedicated protocols with little anatomy in view, the knowledge-base region of an unsegmented organ, and the candidacy of organs outside $\mathcal{H}_C$) fixed errors and estimated no likelihood, prior or threshold; the pilot series are reported separately.

The likelihoods of image grounding are built only from public expert annotations: tumour masks of the R2Seg collections (bladder, breast, cervix, colon, kidney, liver, lung, pancreas, prostate and uterus, including benign uterine fibroids; [TBD] masks) and normal anatomy of TotalSegmentator CT (v3.0, scans without pathology) and MR (v3.0, scans from the original release source; scans indexed from the Imaging Data Commons, which hosts TCIA collections, are excluded to avoid overlap with the evaluation cohort), adults only ([TBD] scans). Neither enters the evaluation cohort.

3.2 Implementation

The tumour agents are BiomedParse (2D) and VoxTell (3D promptable); the anatomy agents are TotalSegmentator and VoxTell. The knowledge base has 204 entities and 433 relations (v0.5.8). Host and tumour reasoning are deterministic set operations. The likelihood tables of image grounding and the outcomes that rule out or confirm a tumor at level α, the compatibility weights β, the parsimony term ψ and the flag factors η_f are estimated on expert data and frozen before any TCGA result (Section 2.3); the remaining design choices (the ordinal support levels, the decision table, the error level α and the geometric constants) and their sources are listed in Supplementary Table S1. The whole cohort was processed on a CPU cluster in [TBD] hours.

3.3 Evaluation without voxel ground truth

Without voxel annotations, we use stage-free yardsticks that a correct segmentation must satisfy. All are computed on the baseline study only:

- *spread*: the median $|\log(v/\tilde{v})|$ of the index-lesion volume v across the series of one study, where $\tilde{v}$ is the study median;
- *cross*: the share of same-study series pairs whose sizes differ by at least one RECIST progression step;
- *disagree*: the share of series whose index lesion is not the study's majority lesion;
- *miss*: the share of baseline series with no measurable lesion (the primary tumour is present before surgery).

In addition, [TBD] agent candidates were rated blind by a reader (tumour, not tumour, unsure); these ratings measure the precision of image evidence against a candidate and the share of rated tumours it rejects. The patient-median Spearman correlation with T and overall stage is reported for reference only. No parameter, rule or threshold is selected on stage or on the ratings.

3.4 Validation of image grounding on expert data

Because TCGA has no tumour annotation, the reasoning is validated where truth is known, on a testbed built from expert data only (R2Seg collections with expert organ and tumour labels: kidney, liver, pancreas, lung, cervix and prostate; and TotalSegmentator normal anatomy). Agents are not run on these data; instead, agent-like candidates are simulated from the expert masks: cores (partial sub-regions and three-slice cuts), outlines (exact, dilated by 3–5 mm, grown into the host's own parenchyma, leaking into a neighbour, organ-sized, and with a false lobe joined by a thin bridge), normal candidates (vessel and bowel tube segments, organ parts and edges, salient intra-organ structures), and perturbed organ masks (leaks, merges, missing parts, the tumour included in the organ, and the true host unsegmented and represented by its knowledge-base region, with the regions of its neighbours as negatives). Every estimate is held out (leave one R2Seg collection out; two folds of TotalSegmentator). We report: (i) existence: for each α, the held-out rates at which expert tumour cores are ruled out and expert normal candidates are confirmed, overall and for candidates below 3 ml; (ii) host: top-1 attribution and the coverage and size of the host credible set of expert tumours among their true and neighbouring organs, and how often the credible set holds both organs of synthetic invasion cases (an expert tumour joined to a band of its labelled neighbour), separately

Table 3: Image-grounded reasoning per cancer type (baseline series). *Reasoning only*: accepted anchors with every outcome neutral. *Rejected/recovered*: anchors whose decision the existence test changed. *Host set*: anchors whose host credible set holds more than one organ. *Override*: primary-not-detected overrides.

Cancer	Series	Reasoning only	Grounded	Rejected	Recovered	Host set > 1	Override
			[TBD]				
All	3,291						

Table 4: Stage-free yardsticks on 3,116 baseline series (942 patients). The Spearman ρ with T stage is reported only and was not used for any choice.

Method	Spread	Cross (%)	Disagree (%)	Miss (%)	ρ_T (reported)
BiomedParse			–		
VoxTell			–		
Reasoning only					
Full model					

for intraparenchymal, straddling and exophytic tumours and under perturbed organ masks; (iii) extent: how often $\hat{\xi}$ is the extent closest to the expert mask, its Dice, its volume ratio, and how often it grows into host parenchyma; (iv) sensitivity to $\alpha \in \{0.01, 0.05, 0.10\}$, to τ and the host tolerance δ, and to the image quality-control thresholds. [TBD]

3.5 Effect of image-grounded reasoning

Host and tumour reasoning are identical with and without image grounding ($\Lambda = 1$), so the comparison isolates it. Table 3 reports, per cancer type, the anchors accepted, uncertain and rejected by the support levels alone (every outcome neutral) and with the existence test; anchors whose host credible set holds more than one organ; overrides of the decision policy; lesions whose host joined the expected list or is unknown; and the volume between the anchor and the union of outlines that the chosen extents keep. [TBD: main pattern.] On the rated candidates, image evidence against a candidate is correct in [TBD] % and rejects [TBD] % of rated tumours.

3.6 Baseline yardsticks

Table 4 compares BiomedParse alone, VoxTell alone, reasoning only, and the full model on the yardsticks of Section 3.3. [TBD: main finding, stated without reference to stage.]

3.7 Qualitative results

[TBD: Fig. 4 to be regenerated with the G0 outputs: an anchor with its outlines and priors, its evidence outcome, test result and likelihood ratio per host in its credible set, and the chosen extent.]

3.8 Ablation

[TBD, on the expert testbed and with the columns of Table 3 on TCGA: (i) numerical support priors ($\pi_{\text{two}} \in \{0.75, 0.9\}$, $\pi_{\text{single}} = 0.5$, odds factor 1/3 per warning signal, thresholds 0.75/0.25) instead of the support levels and the test at α; (ii) the knowledge-base host weights (1/0.3/0.15) instead of equal weights; (iii) each evidence component removed in turn (context, internal structure, geometry); (iv) no parsimony term ψ or flag factors η_f, and the extent evidence added; (v) flagged outlines removed from $\mathcal{X}_a$ instead of penalised; (vi) binary touch instead of the compatibility features; (vii) a ruled-out two-agent anchor rejected instead of uncertain; (viii) $\alpha \in \{0.01, 0.05, 0.10\}$.]

3.9 Limitations

- Image quality control relies on what the anatomy agents can segment; a scan whose organs fail for reasons other than quality (unusual anatomy, post-surgical change outside the primary, or a dedicated protocol with little anatomy in view) may be graded degraded.
- There is no voxel-level ground truth, so accuracy is assessed through stage-free yardsticks, expert data and blind ratings of candidates.

figure to be regenerated

Figure 4: Image-grounded reasoning on a public TCGA scan. Left: anchors (filled), outlines of each agent and the reasoning priors. Middle: evidence state and likelihood ratio of each extent under each host. Right: accepted tumours with their chosen extents. [TBD]

- Image evidence decides existence and chooses among the agents' outlines; it does not redraw boundaries. A tumour that no agent outlines, or outlines only in part, is not recovered beyond the agents' proposals.
- Agent agreement cannot be calibrated on expert data, because the agents are not run there; it enters as ordinal support levels, and existence is tested rather than given a posterior probability.
- On expert data the existence test rules out far more often than it confirms: few outcomes confirm a tumour of 3 ml or more, so acceptance of larger candidates rests mainly on agent agreement.
- The site of a tumour often cannot be decided from the image; the host is reported as a credible set, and the image does not re-rank hosts.
- The extent is chosen by a prior learned on expert data, not by image evidence, which did not improve the choice of extent on held-out expert data.
- Contiguous extension of one tumour into a neighbouring organ is not modelled: anchors within 10 mm share one host. The host testbed contains intraparenchymal expert tumours only.
- The expert three-slice cores of the existence tables are cut on the common ~ 5 mm grid; agent cores of three native slices are thinner, and on the testbed such cores were ruled out in 21% of cases at $\alpha = 0.05$ [to be resolved by rebuilding these items at native resolution].
- Other or indeterminate lesions are reported but not characterised.
- Many agent false positives remain uncertain rather than organ-like: normal sub-structures that resemble tumour (intra-organ vessels, renal sinus, bowel loops) are the main confounder, and iso-intense candidates have no context test.
- Evidence is weaker on MR than on CT, and hosts without expert normal anatomy (for example ovary and breast) use pooled normal tables.
- The default distant-site lists do not include brain, and pleura and distant lymph nodes have no anchor in the knowledge base, so metastases there are not reported.

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Table 5: Supplementary Table S1. Constants of the v7 reasoning and their source: *expert* (estimated on public expert annotations, R2Seg and TotalSegmentator; held out where evaluated), *KB* (knowledge base), *design* (chosen a priori, with the reason; sensitivity reported). No constant is estimated on TCGA or on stage.

Constant	Value	Source	Note
Host weights	equal	design	$\mathcal{H}_C$ decides candidates only; roles attached after inference
Envelope, lesion distance τ	10 mm, 10 mm	design	tumor displacement; fixed in mm (not slice spacing)
Mask confidence μ	1, $\frac{1}{2}$, 0	design	feasible / anatomy violated / corrupt or impossible
Organ checks (volume, side, span, contact)	$2v_{\max}$, $0.5v_{\min}$, 10 mm, span$\pm$(var+1)	KB	lower μ; never remove
Compatibility weights β	fitted (choice model)	expert	testbed host cases incl. knowledge-base regions [to be refitted]
No-host score	α-quantile	expert	of true-host scores of expert tumours [under revision]
Ridge penalties	$\frac{1}{2}\|\beta\|^2$; $0.01\|\theta\|^2$	design	host choice model; extent prior
Host credible set	probability $\geq 1-\alpha$	design (α)	smallest such set of candidates
Knowledge-base region (unsegmented)	own span$\pm$(var+$\frac{1}{2}$) levels	KB	sided, in body; out of view if $< 50\%$ of span in scan
Organs outside $\mathcal{H}_C$ as candidates	$\mu_o = 1$, span in view $\geq 50\%$	design	never a knowledge-base region
Support levels ℓ	2 / 1 / 0	design	ordinal: two agents > one agent; each warning signal -1
Error level α	0.05	design	significance level of the existence test; sensitivity $\{0.01, 0.10\}$
Lobe opening, minimum	3 mm ball, 0.5 ml	design	neck narrower than ~ 6 mm
Claim minimum, crumbs	0.5 ml; 0.2 ml within 5 mm	design	as v6
Prompt mismatch, flatness	$< 50\%$ in envelope; aspect ≤ 0.25, coverage $\geq 40\%$	design	warning signals only
Organ-overlap tests	0.6 / 0.6 / 0.8; 0.8 / 0.5	design	as v6 (D10)
Parsimony ψ	piecewise on $\log(\|\xi\|/\|a\|)$	expert	bins $< 0.05, 0.4, 0.8, 1.2, 1.8, 2.5, \infty$ [to be refitted with η_f]
Flag factors η_f	≤ 1	expert	fitted jointly with ψ (held out by collection); no-example flags take the smallest
Extent evidence	not used	expert	did not improve extent choice on held-out expert data (ablation)
Normal reference	2 mm margin, 200 voxels in 50 mm	design	as G0
Sign thresholds (closed, open, patchy, elongated, deviation)	1.42, 5.27, 1.047, 0.363, 1.5	expert	percentiles of expert data (frozen 12 Oct)
Tables (strata, smoothing)	≥ 30; $\kappa = 0.2$	expert	κ by held-out likelihood over $\{0, \ldots, 5\}$; $n/(n+10)$ for reported Λ only
Common grid	~ 5 mm through-plane	expert	thickening changes signs; small-lesion check
Decision	Table 2	design	a conflict gives uncertain
Quality control	$f \geq 0.25$, 0.5; 8 mm; 76.6 HU, SNR 1.64	expert/design	noise percentiles of expert data

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